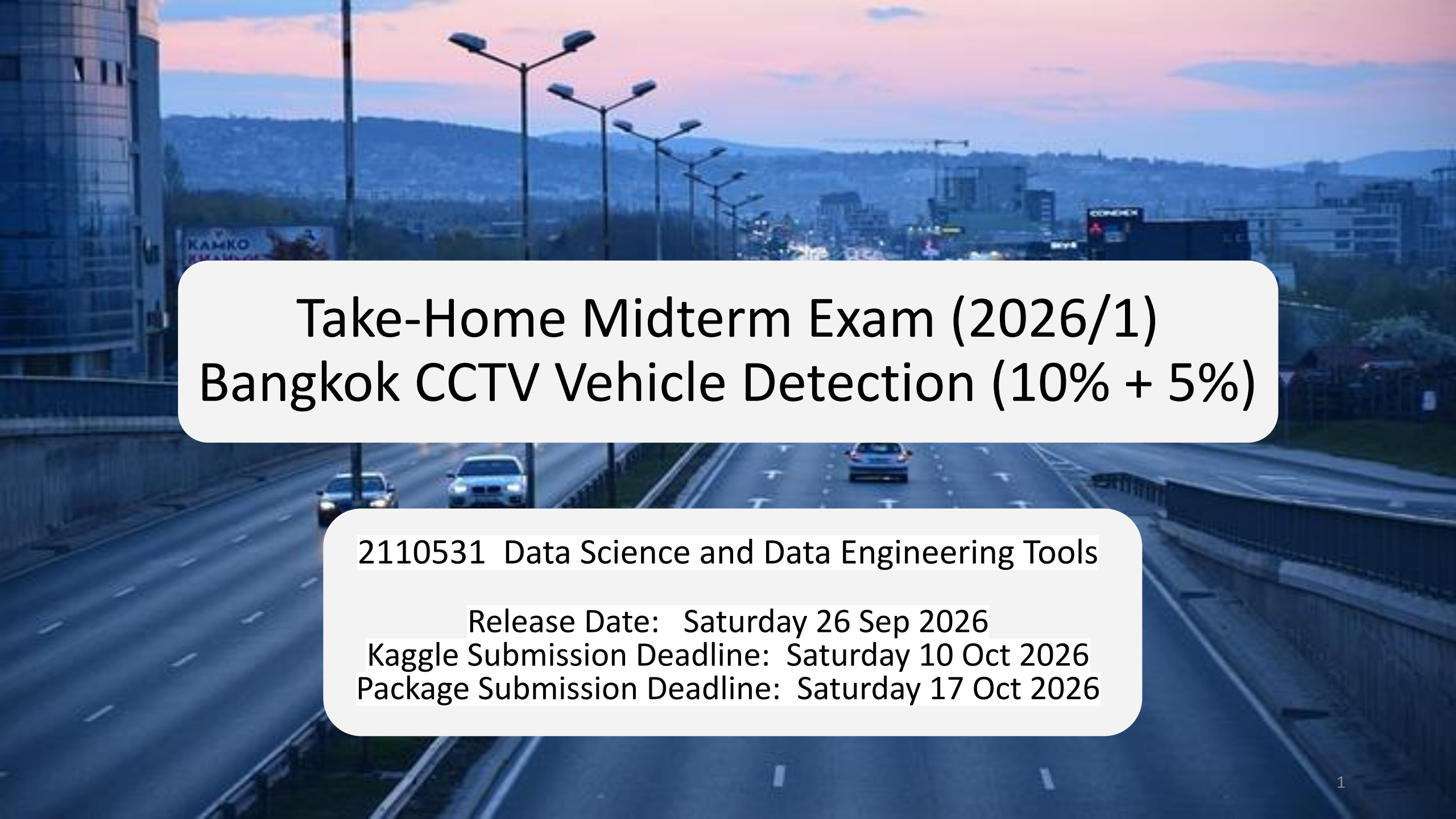




Take-Home Midterm Exam (2026/1) Bangkok CCTV Vehicle Detection (10% + 5%)

2110531 Data Science and Data Engineering Tools

Release Date: Saturday 26 Sep 2026

Kaggle Submission Deadline: Saturday 10 Oct 2026

Package Submission Deadline: Saturday 17 Oct 2026

Introduction:

Data Source

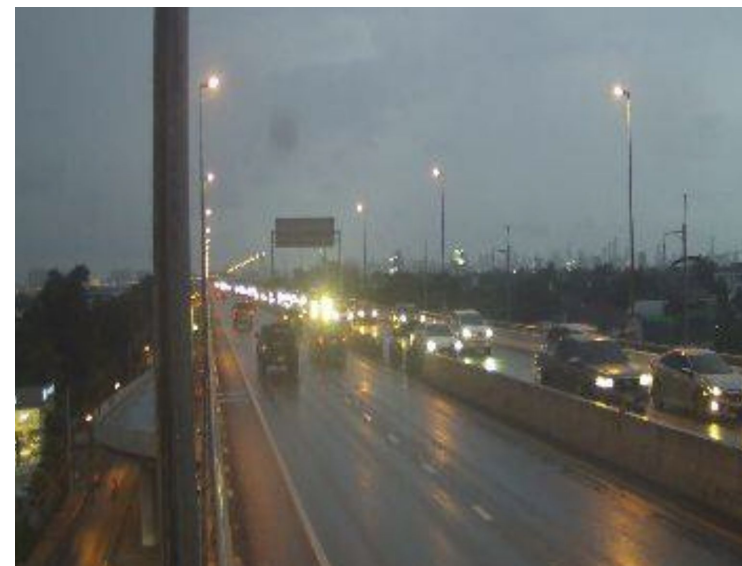
- Traffic camera data scraped from the **Bangkok Metropolitan Administration (BMA) Traffic** platform (<https://cpudapp.bangkok.go.th/bmatraffic/>)
- Data collected automatically at **30-minute intervals**

Dataset Coverage

- Initially collected from a large number of traffic cameras
- After data quality screening, **20 cameras** were selected

Dataset Characteristics

- Traffic camera videos capturing **real-world road conditions**
- Provides visual information of vehicles and other road objects
- Suitable for **computer vision tasks**, particularly object detection



Objective: Object Detection

- Be able to perform data preprocessing & clean data
- Be able to perform a deep learning model (model development & model evaluation)



Example of object detection results: Cars, tuk-tuks, pickup trucks, trucks, and motorcycles

Dataset

Object Detection — 8 classes

20 Cameras from 25/08/2026 06:00 - 31/08/2026 21:00 (every 30 minutes)

- Train 2,991 images
- Test 1,013 images

ID	Class	Details (English)	Details (Thai)
0	Car	Passenger car, including 4-door, 5-door, and standard SUVs	รถเก๋ง 4 ประตู, 5 ประตู, SUV ทัวไป
1	Motorcycle	Motorcycle; a single bounding box includes the rider and passenger(s)	มอเตอร์ไซค์ ตีกรอบเดียวรวมคนขับ + คนซ้อนท้าย
2	Bus	Bus, coach, or large passenger vehicle	รถเมล์ รถทัวร์ รถโดยสารขนาดใหญ่
3	Truck	Semi-truck, trailer truck, dump truck, container truck, or trucks with 6 wheels or more	สิบล้อ, รถพ่วง, รถหัวลาก, รถตู้ทึบ, รถบรรทุก 6 ล้อขึ้นไป
4	Tuktuk	Three-wheeled motorized passenger vehicle (Tuk-Tuk)	รถสามล้อเครื่องรับจ้าง (ตุ๊กตุ๊ก)
5	Van	Passenger van	รถตู้โดยสาร
6	Pickup	Standard pickup truck for personal use	กระบะส่วนบุคคลทั่วไป
7	Songthaew	Modified pickup truck or small truck with two-row passenger seating in the rear	กระบะ/รถบรรทุกเล็กดัดแปลงมีที่นั่ง 2 ฝั่งรับส่งผู้โดยสาร รถสองแถว

Platform: Kaggle

Please join Kaggle competition ASAP:

<https://www.kaggle.com/t/4ba53aa63cc2424c86eb156f71b837b1>

2110531 DSDE (2026/1)

Take-Home Midterm Exam - Bangkok CCTV Vehicle Detection



Dataset Download :

2110531 DSDE (2026/1)

Settings Overview **Data** Discussion Leaderboard Rules Team

train.csv (1.39 MB)

Detail Compact Column

About this file

This file does not have a description yet.

id	image_id	class_id	confidence	x1	y1
0	1066_20260825_060110.jpg	0	1.0	152	121
1	1066_20260825_060110	0	1.0	119	71

Data Explorer

158.11 MB

- test
- train
- sample_submission.csv
- train.csv

Summary

- 4006 files
- 16 columns

Download All

+ New Version

DOWNLOAD VIA

kagglehub

New to Kaggle API? Here's how to [set up your API keys](#).

```
import kagglehub

# Download latest version
path = kagglehub.competition_download('2110531-dsde-2026-1')

print("Path to competition files:", path)
```

Download as zip

This is an individual take-home midterm exam.

Scoring

- 10% for the Kaggle result
- 5% for the submission package

Evaluation Metric: mAP@50

mAP@50 (mean Average Precision at IoU = 0.50) is a standard evaluation metric for object detection.

A prediction is considered a **True Positive (TP)** when it has the correct class and matches a ground-truth box with **IoU ≥ 0.50**.

Average Precision (AP) is calculated from the Precision–Recall curve for each class. The final mAP@50 score is the mean AP across all classes.

AP is calculated using COCO-style evaluation via the `pycocotools` library to ensure standardized and consistent scoring across all submissions.

Predictions + Ground Truth

Rank predictions by confidence

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}$$

IoU matching (same image and class, IoU ≥ 0.50)
Each ground-truth box can be matched only once.

TP / FP / FN

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

TP (Correct Detection),
FP (False Detection – Wrong Class or
IoU < 0.50 or duplicate),
FN (Missed Detection)

Vary confidence threshold

Precision–Recall curve for each class
(IoU threshold = 0.50)

Calculate

AP50 for class c = Area under the
PR curve

Average AP50 across all classes

$$\text{mAP}_{50} = \frac{1}{C} \sum_{c=1}^C \text{AP}_{50,c}$$

Kaggle Scoring Criteria (10%)

There will also be a winner for the technical aspect regardless of the rank (full 10 points).

- 5% from your mAP@50
 - mAP ≥ 0.56 : 5 points
 - mAP ≥ 0.53 : 4 points
 - mAP ≥ 0.50 : 3 points
 - mAP ≥ 0.47 : 2 points
 - mAP ≥ 0.44 : 1 point
 - Otherwise: 0 points

- 5% from your Percentile
 - Percentile [1 - 30] : 5.0 points
 - Percentile [31 - 60] : 4.5 points
 - Percentile [61 - 100] : 4.0 points

Important Rules!!!

1) The result must be able to replicated by the code you submitted. Your code must be able to run your TA's. Otherwise, your project can't be graded.

2) Copying is prohibited! 

3) Can we use VLM? Yes

Submission+Report (5%)

- Take-home midterm exam report:
 - Part 1: (1) source code, (2) uploaded csv (result), (3) your prepared data, (4) model weight
 - Please note that your result must be able to be reproduced, and it **must be similar (close) to the result on Kaggle.**
 - Part 2: Report (word & pdf)
 - Chapter 1: Introduction
 - Chapter 2: Data preparation (How to prepare data along with data statistics)
 - Chapter 3: Model (explanation of your model)
 - If you call the model API (Gen AI), you need to understand about the model and explain it in the report (e.g., model name, model size, pretrained data, etc.) along with **cost**!
 - Likewise, for the DL model, you need to give details of the DL model that you have selected and trained (e.g., model architecture, etc.).
 - Chapter 4: Results (must contain captured screen on Kaggle)
 - Chapter 5: Discussion, e.g., error analysis and how to improve it
 - Chapter 6: Conclusion
- Submission:
 - MCV (Take-home midterm exam: report submission)
 - **Package Submission Deadline:** Saturday 17 Oct 2026
 - Please be careful before submission, as Kaggle allows no more than **5** submissions per day.
- **If we cannot access your Google Drive link or GitHub link, it is NOT possible to grade your work, resulting in no score.**
- **You are not allowed to modify your Google Drive or GitHub after the submission date.**

Hint: You can find example codes in the course GitHub here.

Week 05: Intro to Deep Learning

1. Image classification with CNN [PyTorch Lightning] (~10 min):  [Open in Colab](#)
2. Image classification with Pretrained Model
 - 2-1. Image classification with EfficientNetV2s [PyTorch Lightning] (~3 min):  [Open in Colab](#)
 - 2-2. Image classification with EfficientNetb0 (Load a Pretrained Model from Hugging Face) [PyTorch Lightning] with Weights & Biases (~3 min):  [Open in Colab](#)
3. Object detection with YOLOv8
 - 3-1. Object detection with YOLOv8 (basic script) [PyTorch Lightning] (~20-30 min):  [Open in Colab](#)
 - 3-2. Object detection with YOLOv8 (custom dataset) [PyTorch Lightning] (~10 min):  [Open in Colab](#)
4. Semantic segmentation with deeplabv3 [PyTorch Lightning] (~15 min):  [Open in Colab](#)
5. Image classification with ViT from Hugging Face & TensorBoard [PyTorch] (~10 min):  [Open in Colab](#)